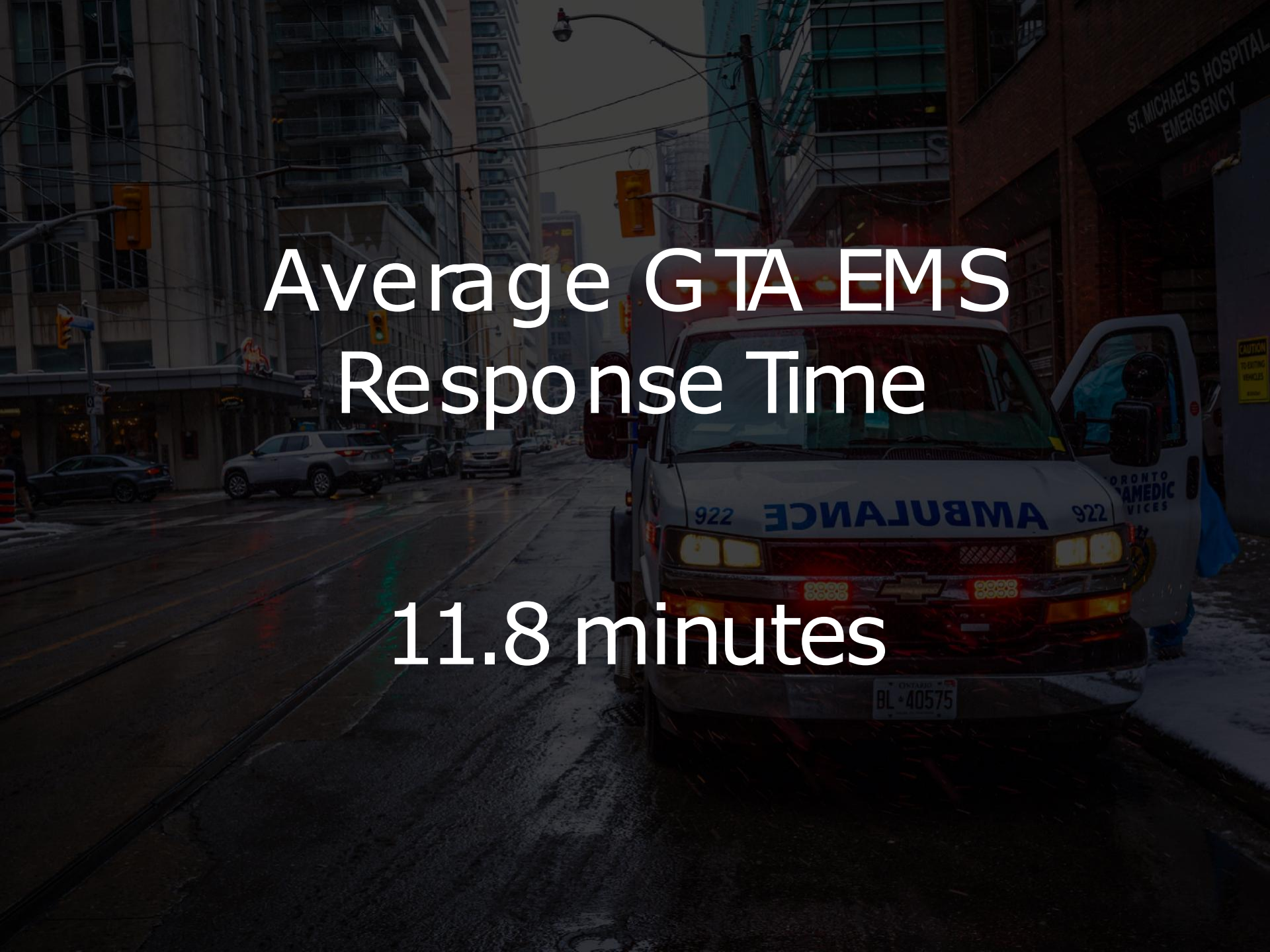
UNIVERSITY OF TORONTO
FACULTY OF DENTISTRY

MEDICAL EMERGENCIES IN THE DENTAL OFFICE



Average GTA EMS Response Time

11.8 minutes

TIMING OF EMERGENCIES

Anesth Prog. 1989 Jul-Oct;36(4-5):223-5.

Analysis of systemic complications and deaths during dental treatment in Japan

H Matsuura

Timing	%
Pre-treatment	1.5
Local anesthetic delivery	54.9
Peri-treatment	22.9
Post-treatment (Office)	15.2
Post-treatment (Home)	5.5

TIMING OF EMERGENCIES

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Analysis of systemic complications and deaths during dental treatment in Japan

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Treatment	%
Extractions	38.9
Endodontics	26.9
Tooth preparation	7.3
Intraoral incisions	1.7
Prosthodontics	0.3
Apicoectomy	0.7
Restorative	2.3
Removal of restorative material	0.7
Other treatment	9.0
Unknown	12.3

DEATH IN THE OFFICE

- Rare
  - 2.6 per year
- Factors
  - Medications
  - Providers
  - Age
  - Setting

Review > Oral Surg Oral Med Oral Pathol Oral Radiol. 2017 Feb;123(2):194-204.e10.
doi: 10.1016/j.oooo.2016.10.015. Epub 2016 Nov 2.

Death related to dental treatment: a systematic review

Nathan G Reuter [1], Philip M Westgate [2], Mark Ingram [3], Craig S Miller [4]

	Airway/respiratory complications	*Local or systemic bleeding*	*Cardiovascular events*	*Anesthesia/sedation/ medication-related complications*	*Infection*	*Other causes*	*Total deaths*
1960–1969	0	0	0	2	1	0	3
1970–1979	4	1	15	40	0	0	60
1980–1989	0	1	4	9	0	1	15
1990–1999	6	1	5	10	3	2	27
2000–2009	5	1	4	1	4	1	16
2010–2015	3	1	3	8	11	1	27
							148

MITIGATING FACTORS

- Factors that increase the likelihood of emergencies
  - Advanced age
  - Therapeutic medical advances
  - Longer appointments
  - Drug administration

DEFINITION OF EMERGENCY

- Failure of compensation in face of stress

MEDICALLY COMPROMISED PATIENTS

- Limited respond to stress
- Decompensate sooner when faced with stress

ASA PHYSCIAL STATUS CLASSIFICATION SYSTEM

ASA PS Classification	Definition	Adult Examples, Including, but not Limited to:	Pediatric Examples, Including but not Limited to:	Obstetric Examples, Including but not Limited to:
ASA I	A normal healthy patient	Healthy, non-smoking, no or minimal alcohol use	Healthy (no acute or chronic disease), normal BMI percentile for age	
ASA II	A patient with mild systemic disease	Mild diseases only without substantive functional limitations. Current smoker, social alcohol drinker, pregnancy, obesity (30<BMI<40), well-controlled DM/HTN, mild lung disease	Asymptomatic congenital cardiac disease, well controlled dysrhythmias, asthma without exacerbation, well controlled epilepsy, non-insulin dependent diabetes mellitus, abnormal BMI percentile for age, mild/moderate OSA, oncologic state in remission, autism with mild limitations	Normal pregnancy*, well controlled gestational HTN, controlled preeclampsia without severe features, diet-controlled gestational DM.
ASA III	A patient with severe systemic disease	Substantive functional limitations; One or more moderate to severe diseases. Poorly controlled DM or HTN, COPD, morbid obesity (BMI ≥40), active hepatitis, alcohol dependence or abuse, implanted pacemaker, moderate reduction of ejection fraction, ESRD undergoing regularly scheduled dialysis, history (>3 months) of MI, CVA, TIA, or CAD/stents.	Uncorrected stable congenital cardiac abnormality, asthma with exacerbation, poorly controlled epilepsy, insulin dependent diabetes mellitus, morbid obesity, malnutrition, severe OSA, oncologic state, renal failure, muscular dystrophy, cystic fibrosis, history of organ transplantation, brain/spinal cord malformation, symptomatic hydrocephalus, premature infant PCA <60 weeks, autism with severe limitations, metabolic disease, difficult airway, long term parenteral nutrition. Full term infants <6 weeks of age.	Preeclampsia with severe features, gestational DM with complications or high insulin requirements, a thrombophilic disease requiring anticoagulation.
ASA IV	A patient with severe systemic disease that is a constant threat to life	Recent (<3 months) MI, CVA, TIA or CAD/stents, ongoing cardiac ischemia or severe valve dysfunction, severe reduction of ejection fraction, shock, sepsis, DIC, ARD or ESRD not undergoing regularly scheduled dialysis	Symptomatic congenital cardiac abnormality, congestive heart failure, active sequelae of prematurity, acute hypoxic-ischemic encephalopathy, shock, sepsis, disseminated intravascular coagulation, automatic implantable cardioverter-defibrillator, ventilator dependence, endocrinopathy, severe trauma, severe respiratory distress, advanced oncologic state.	Preeclampsia with severe features complicated by HELLP or other adverse event, peripartum cardiomyopathy with EF <40, uncorrected/decompensated heart disease, acquired or congenital.
ASA V	A moribund patient who is not expected to survive without the operation	Ruptured abdominal/thoracic aneurysm, massive trauma, intracranial bleed with mass effect, ischemic bowel in the face of significant cardiac pathology or multiple organ/system dysfunction	Massive trauma, intracranial hemorrhage with mass effect, patient requiring ECMO, respiratory failure or arrest, malignant hypertension, decompensated congestive heart failure, hepatic encephalopathy, ischemic bowel or multiple organ/system dysfunction.	Uterine rupture.
ASA VI	A declared brain-dead patient whose organs are being removed for donor purposes			

EMERGENCY SPECTRUM

Emergency	n
Syncope	15,407
Mild allergic reaction	2583
Angina pectoris	2552
Postural hypotension	2475
Seizures	1595
Asthmatic attack (bronchospasm)	1392
Hyperventilation	1326
"Epinephrine reaction"	913
Insulin shock (hypoglycemia)	890
Cardiac arrest	331
Anaphylactic reaction	304
Myocardial infarction	289
Local anesthetic overdose	204
Acute pulmonary edema (heart failure)	141
Diabetic coma	109
Cerebrovascular accident	68
Adrenal insufficiency	25
Thyroid storm	4
TOTAL	**30,608**

> J Am Dent Assoc. 1986 Apr;112(4):499-501. doi: 10.14219/jada.archive.1986.0043.

Emergency preparedness: a survey of dental practitioners

T B Fast, M D Martin, T M Ellis

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Managing medical emergencies

S F Malamed [1]

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Managing medical emergencies

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CONSIDERATIONS

- Approach
  - Prevention
  - Preparation
  - Recognition
  - Action
- Awareness
  - Patient factors
  - Signs and symptoms of common conditions
  - Proper responses to emergencies
  - Pathophysiology of common diseases

CHECKLIST

- Assigned roles
- Contingency plan
- Training
- Emergency equipment
- Emergency drugs
- Drills
- Emergency numbers

BASIC EMERGENCY EQUIPMENT

- Automatic external defibrillator
- Yankauer suction
- Syringes and needles
- Bag valve mask
- Oxygen
- Portable oxygen system
- Glucometer
- Automated sphygmomanometers
- Stethoscope
- Pulse oximeter
- Valved holding chamber

EMERGENCY DRUGS

Royal College of
Dental Surgeons of Ontario

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Preparing for a medical emergency in the dental office

Drug	Indication	Initial Adult Dose	Recommended Pediatric Dose
Oxygen	Most medical emergencies	100% inhalation	100% inhalation
Epinephrine	Anaphylaxis	0.3-0.5 mg IM or 0.01-0.1mg IV	0.01 mg/kg
	Bronchospasm	0.3-0.5 mg IM or 0.01-0.1 mg IV	0.01 mg/kg
	Cardiac Arrest	1mg IV	0.01 mg/kg
Nitroglycerin	Angina pectoris	0.3 mg or 0.4 mg SL	Not indicated
Diphenhydramine	Allergic reactions	50mg IM or IV	1 mg/kg
Salbutamol	Asthmatic bronchospasm	2 puffs (100 mcg/puff)	1 puff
ASA (non-enteric coated)	Acute myocardial infarction	160 to 325 mg PO	Not indicated

OTHER RECOMMENDED DRUGS

- Glucose
- Naloxone
- Flumazenil

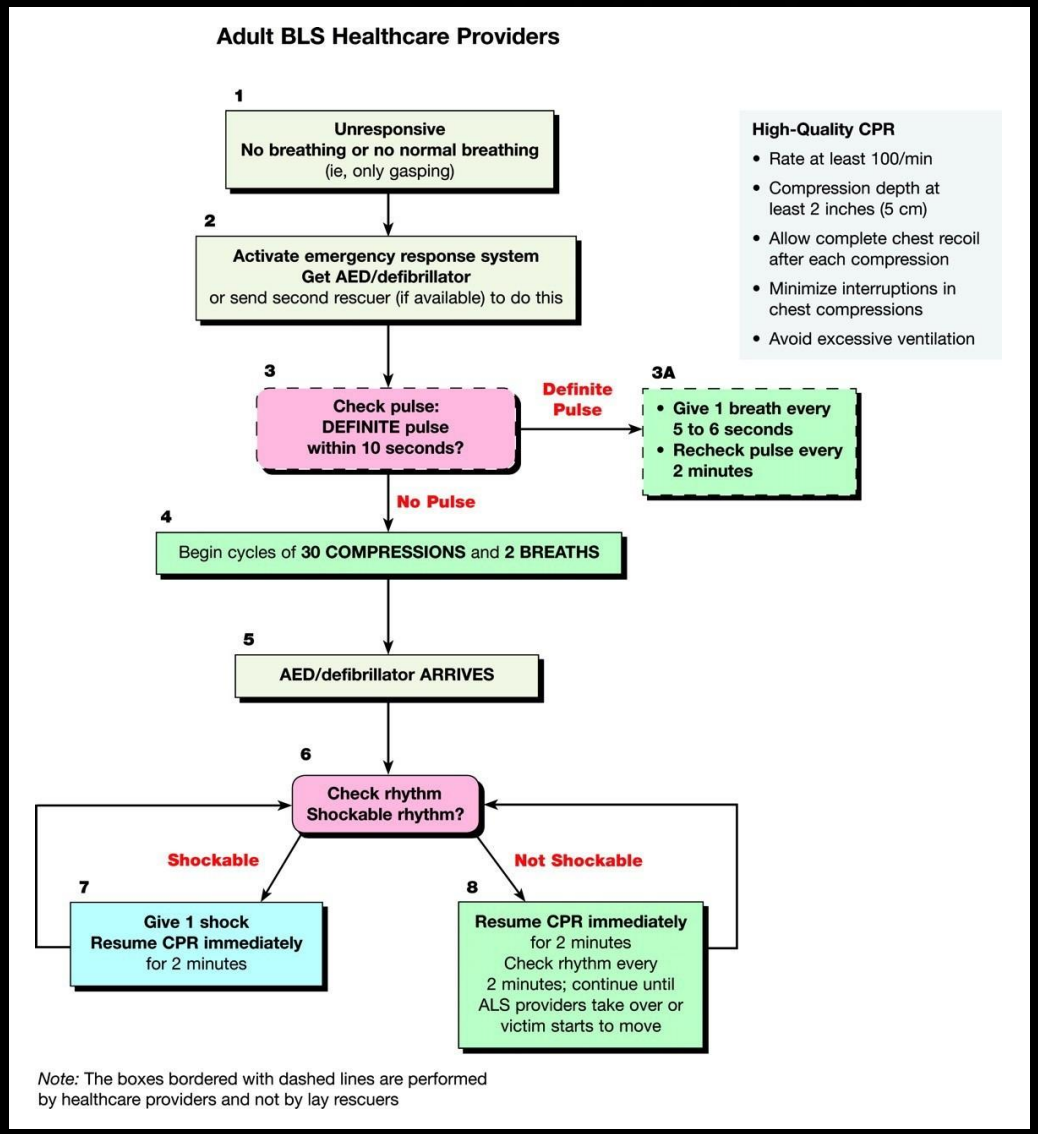
BASIC APPROACH

- Discontinue treatment
- Assess the level of consciousness
- Activate the office emergency system
- Position the patient
- Assess airway, breathing and circulation
- Monitor vital signs
- Reassess
- Follow-up

CARDIAC ARREST

- Signs and Symptoms
  - Apnea
  - Cyanosis
  - Unresponsiveness
- Vitals
  - No pulse
- Cause
  - Cardiac pathology

CARDIAC ARREST



CARDIAC ARREST

- General
  - Call for help, activate EMS, get AED
- Circulation
  - Check pulse for no more than 10 seconds
  - Start compressions
  - One operator: 30 compressions per 2 breaths, rate of 100 compressions/minute
- Airway
  - Head tilt-chin lift, jaw thrust
  - Examine airway, suction mouth
- Breathing
  - Watch for chest rise
- Defibrillation
  - Attach and use as soon as it is available
  - Shock if indicated and repeat analysis every 2 minutes
  - Resume compressions immediately after shock

CHEST PAIN

- Signs and Symptoms
  - Substernal pain that could radiate to arms, neck, jaw or abdomen
  - Myocardial infarction usually more severe symptoms than angina
- Cause
  - Inadequate blood flow to myocardium

CHEST PAIN

- Consider if previous history of angina (stable or unstable)
- Differential diagnosis
  - Gastrointestinal
  - Musculoskeletal
  - Psychological
  - Respiratory

ANGINA PECTORIS

- Past medical history
  - Hypertension
  - History of angina
  - History of MI
  - History of stroke
  - Diabetes
  - Cholesterol
- Obesity
- Age
- Male
- Family history

ANGINA PECTORIS

- Supportive
  - Vitals
  - Oxygen
- History of angina present
  - Administer nitroglycerin
    - Caution for SBP < 90 mmHg or administration of 5-PDE inhibitors
  - No resolution
    - Activate EMS, non-EC ASA 160 mg
- No history of angina present:
  - EMS
  - Vitals
  - Oxygen
  - Aspirin
  - Consider nitroglycerin

SYNCOPE

- Signs and Symptoms
  - Sweating, nausea, anxiety, loss of consciousness
- Vitals
  - Bradycardia, hypotension
- Cause
  - Variable
    - Cardiogenic versus non-cardiogenic
- Treatment
  - Vitals
  - Supine positioning
  - ABCs
  - Oxygen
  - Cold compress

HYPOGLYCEMIA

- Signs and Symptoms
  - Hunger, weakness, sweating, confusion, unresponsiveness
- Vitals
  - Tachycardia, hypotension, hypoglycemia
- Cause
  - Lack of blood glucose to the brain, history of diabetes (usually insulin dependent)
- Treatment
  - Position
    - Conscious patient: Upright
    - Unconscious: Supine
  - ABCs
  - Vitals
  - Drugs:
    - Glucose if conscious
    - Oxygen and glucagon or IV D50W if available
  - Activate EMS

ADRENAL INSUFFICIENCY

- Signs and Symptoms
  - Confusion, weakness, vomiting
- Vitals
  - Hypotension
- Cause
  - Adrenal suppression (primary or secondary)
- Treatment
  - Supine positioning or Trendelenberg
  - ABCs
  - Vitals
  - Drugs
    - Steroids
    - Oxygen
  - Call EMS if necessary

STROKE

- Signs and Symptoms
  - Dizziness, vomiting, unilateral, weakness
- Vitals
  - Often normal
- Cause
  - Brain ischemia due to thrombosis, embolism, or systemic hypoperfusion
- Treatment
  - Reclined position with elevation of head
  - ABCs
  - Vitals
  - Oxygen
  - Activate EMS

SEIZURE

- Signs and Symptoms
  - Confusion, rolling back of eyes, loss of consciousness, tonic phase (clenching of teeth), clonic phase (tremors, convulsions)
- Causes
  - Syncope, drug reactions, hypoglycemia, hyperventilation, CVA, seizure disorder
- Treatment
  - Supine positioning
    - Protect patient from injury and clear instruments
  - After convulsions: ABCs, Oxygen, Vitals
  - Status epilepticus: Call EMS

LOCAL ANESTHETIC SYSTEMIC TOXICITY (LAST)

- Signs and Symptoms
  - Confusion, restlessness, convulsions, lightheadedness
- Vitals
  - May show increased BP and HR
- Treatment
  - ABCs
  - Vitals
  - Perform CPR if cardiac arrest
  - Call EMS

ASPIRATION

- Signs and Symptoms:=
  - Coughing/gagging, inability to speak, cyanosis, witnessed object aspiration
- Vitals
  - Monitor breathing, oxygen saturation
- Treatment
  - Conscious
    - Encourage coughing
    - Abdominal thrusts
  - Unconscious
    - CPR
    - Call EMS
    - Emergency surgical airway

ASTHMA

- Signs and Symptoms
  - Sense of suffocation, pressure in chest, expiratory wheeze, prolonged expiratory phase
- Cause
  - Can be induced by allergy, infection, exercise. Usually history of asthma.
- Treatment
  - Upright comfortable position
  - ABCs
  - Vitals
  - Conscious
    - Salbutamol 2.5 to 5.0mg q20min x 3
  - Unconscious
    - 0.3 to 0.5 mg epinephrine 1:1000 IM q20min x 3
  - Call EMS if indicated

ALLERGIC REACTIONS

- Signs and Symptoms
  - Mild pruritis (itching), mild urticaria (rash)
- Vitals
  - Generally normal
- Cause
  - Exposure to allergen
- Treatment
  - ABCs
  - Vitals
  - Drugs
    - Diphenhydramine 25-50 mg IM or PO
  - Consider referral for allergy testing

ANAPHYLAXIS

- Signs and Symptoms:
  - Severe pruritis and urticaria, swelling of lips/eyelids, wheezing, hoarseness, loss of consciousness
- Vitals
  - Hypotension, tachycardia
- Treatment
  - ABCs
  - Vitals
  - Drugs
    - Oxygen
    - 0.3-0.5 mL epinephrine 1:1000 IM
    - Benadryl 25-50 mg SC
    - May consider steroid
  - CPR if cardiac arrest
  - Call EMS

REALITIES FOR EMERGENCIES

- 50% will be syncope
- Potentially life threatening
- Occur during or after local anesthetic

SUMMARY

- Preparedness is key
  - Ensure that you and your staff are up to date with your training
  - Practice mock emergency drills with staff
  - Consider producing cheat sheets for common emergencies
  - Ensure emergency drugs are not expired
  - Ensure all necessary equipment is in good working order